



SILIFOOD: a tool to support the risk assessment of non-evaluated food contact material substances

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ABSTRACT

The SILIFOOD tool was created to support the safety assessment of food contact material (FCM) substances. By combining the retrieval of existing toxicological data with (quantitative) structure-activity relationship [(Q)SAR] predictions, the tool is particularly relevant for FCM substances with limited or no toxicological data like non-intentionally added substances (NIAS) and substances migrating from non-plastic FCM. Existing databases and lists of substances containing toxicological information were identified and curated. In addition to an in-house database with FCM-related information, the European Food Safety Authority (EFSA) OpenFoodTox database, the European Chemicals Agency (ECHA) lists of evaluated substances as well as lists of compounds with toxicological concerns were chosen. (Q)SAR models predicting toxicological endpoints relevant for FCM were selected from VEGAHUB. The endpoints of interest were bioavailability/bioaccumulation potential, genotoxicity, carcinogenicity, developmental and reproductive toxicity, and sub-chronic toxicity. Moreover, models for endocrine activity and Cramer class identification were included. Next, a stand-alone and publicly available software was developed to automatically collect information from the selected lists and databases and to combine them with (Q)SAR predictions. The SILIFOOD tool, developed specifically for single organic FCM substances, was evaluated using case studies.

1. Introduction

Many substances from food contact materials (FCM) are known to migrate into food and, therefore, FCM can be a source of chemical exposure for the consumer (Geueke et al., 2022). European Regulation (EC) No 1935/2004 requires that all materials and articles intended to come into contact directly or indirectly with food must be safe for human health (Regulation (EC) 2021). For plastic FCM, an additional specific regulation exists, i.e., Regulation (EU) No 10/2011, including a positive list of substances authorized for use as starting products (Regulation (EU) No 10/2011). In contrast, no harmonized EU legislation has been implemented for non-plastic FCM. Nevertheless, general recommendations are available for some non-plastic FCM types, including inventory lists of substances known to be used by Member States or national legislations. In practice, the scope of national

legislations varies widely across Member States and is overall rather limited. Consequently, most of the substances used in non-plastic FCM have not, or have not recently, been evaluated for their safety and thus constitute an important source of potential food toxicants. However, the FCM contamination problem is not limited to the migration of non-evaluated starting substances from non-plastic FCM into the food. Both for plastic and non-plastic FCM, migration of non-intentionally added substances (NIAS), including impurities, oligomers, degradation products, and neoformed compounds into the food has also been reported (Muncke, 2011; Nerin et al., 2013). NIAS typically represent the more significant part of the migrants, and their toxicological profile is mostly not or only poorly characterized (Grob, 2014). Furthermore, the legislative framework for these NIAS is limited. For plastic FCM, Regulation (EU) No 10/2011 states that the manufacturer should assess any potential health risk associated with NIAS in accordance with

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internationally recognized scientific principles. For non-plastic FCM, no general authorization or listing exists for NIAS. In 2015, the International Life Sciences Institute (ILSI) published a range of recommendations and a guideline to assess the safety of NIAS in all FCMs (Koster et al., 2015).

Moreover, in 2019, the European Food Safety Authority (EFSA) developed the “Rapid Assessment of Contaminant Exposure (RACE) tool, a risk evaluation methodology to allow a rapid and consistent risk-based classification of notifications through the Rapid Alert System for Food and Feed (RASFF) (Fürst et al., 2019). This RACE tool aimed to provide a transparent set of criteria that can support the evaluation of food contaminants, including those arising from FCM. The RACE tool uses food consumption information from the EFSA Comprehensive European Food Consumption Database to estimate acute and chronic exposure to individual foods and compare the results to the relevant toxicological reference points (RPs) (when available). In practice, the EFSA RACE methodology comprises a set of decision trees that guide the fast risk assessment of food-related compounds, including FCM substances. When a legal limit applies, this value should be used for the risk assessment. However, in cases when there is no legal limit and a health-based guidance value or reference point is available, the risk assessment essentially consists of comparing exposure estimates with these values. If no legal limits nor HBGV or RP are available, alternative strategies are described, such as the application of the threshold of toxicological concern (TTC) approach. However, applying the methodology to non-evaluated FCM substances is challenging, especially when only limited toxicological data are available. In such cases, it is recommended to resort to *in silico* tools like (Q)SAR models to derive/predict toxicological information such as genotoxic or carcinogenic properties (Fürst et al., 2019). (Q)SAR models are based on the hypothesis that a chemical's structure can be linked to its biological (or toxicological) activity, a concept already introduced more than fifty years ago (Selassie, 2003). However, over time, the field of (Q)SARs has evolved tremendously and remains very active due to, among others, its significant potential role in chemical risk assessment. The use of (Q)SAR models in risk assessment to assess genotoxic and carcinogenic properties was formalized for drug impurities in the ICH M7 guideline (International Conference on Harmonization, 2018). This guideline requires two prediction methodologies, one expert rule-based and one statistical-based (QSAR) to maximize sensitivity (Honma et al., 2019). Similarly, the EFSA also recommends the use of at least two independent (Q)SAR models based on different training sets or algorithms (knowledge-based and statistically-based models) (EFSA Scientific Committee et al., 2019). In addition, both guidelines on the use of *in silico* models for regulatory purposes ask for predictions within the applicability domain of the model (Benigni et al., 2019; Fioravanzo et al., 2012). The applicability domain (AD) defining the chemical space for which the model's predictions can be reliable. Combining (Q)SAR models with literature data could facilitate the safety assessment of non-evaluated FCM compounds in a cheap and animal-free manner.

Specific attention should also be given to the endocrine-disrupting (ED) properties of non-evaluated FCM substances, as multiple publications have already linked FCM to ED compounds (EDCs) exposure. In 2009, FCMs were identified as an important potential source of exposure to EDCs (Muncke, 2009). Later work showed that at least 117 known or suspected EDCs are intentionally used to manufacture FCM in the EU and/or the US (Groh et al., 2021). Also, previous results of our group suggest that FCM can be a source of EDCs, as multiple EDCs were shown to migrate from polycarbonate replacement baby bottles (Mertens et al., 2016).

Overall, although the methodology of the RACE tool is useful for evaluating non-evaluated substances such as non-plastic FCM compounds and/or NIAS, the data collection needed to support its application is time-consuming. Moreover, the use of *in silico* tools to fill data gaps is not considered in the RACE tool and the ED potential of substances is not specifically addressed. In this context, the SILIFOOD tool

was developed integrating existing regulatory and toxicological data with *in silico* predictions to ensure a more streamlined and faster risk assessment of non-evaluated FCM substances. Afterward, the applicability of the SILIFOOD tool was investigated by performing four case studies with non-evaluated substances migrating from FCM (non-plastic FCM substances or NIAS).

2. Materials and methods

2.1. Development of the tool

2.1.1. Selection and curation of lists and databases

The EFSA RACE tool (Fürst et al., 2019) was used as a starting point to select relevant toxicological data sources to develop the tool. In line with this methodology, information sources related to genotoxicity, carcinogenicity, health-based guidance values (HBGV) or RPs were identified. In addition, other data sources providing information on the evaluation status and migration limits for FCM were selected. Lists and databases containing information about the ED potential of chemicals were also included. For selecting the relevant information sources, focus was put on (i) the type of data present, (ii) the accessibility of the databases/lists, and (iii) the possibility of integrating them into an automated tool. Finally, databases/lists undergoing regular updates were preferred to ensure up-to-date information.

For all substances included in the selected lists and databases, Simplified Molecular Input Lines Entry System (SMILES) (Weininger, 1988) were retrieved and/or curated to ensure correct identification of the substance of interest and to allow the application of the VEGA prediction models. For lists and databases lacking SMILES, SMILES were generated and curated with a freely available semi-automated workflow implemented in KNIME (Gadaleta et al., 2018). The SMILES curation implies the exclusion of mixtures, reaction masses/products, polymers, isomers, biological materials, inorganic and organometallic compounds, neutralization of salts, removal of characters encoding stereoisomerism, and conversion of the neutralized SMILES into a standardized QSAR-ready format.

2.1.2. *In silico*/(Q)SAR models

(Q)SAR models predicting several relevant toxicological endpoints were identified from VEGAHUB (<https://www.vegahub.eu/>). The VEGA models, developed by Istituto di Ricerche Farmacologiche Mario Negri IRCCS (IRFMN) and their scientific partners, are freely available. The endpoints of interest were defined based on the toxicological data required for the safety assessment of plastic FCM substances. They included genotoxicity, carcinogenicity, reproductive and developmental toxicity, subchronic oral toxicity (90-day study with a No Observed Adverse Effect Level (NOAEL) determination), and bioaccumulation potential (EFSA CEF Panel et al., 2008). Furthermore, models were selected to predict endocrine activity and to assign compounds to their respective Cramer class to enable the application of the Threshold of Toxicological Concern (TTC). All predictions obtained with the VEGA models are associated with an indication of their reliability based on either the applicability domain index (ADI) or the consensus score. The ADI is a global index ranging from 0 (worst case) to 1 (best case) calculated by an algorithm taking into account several parameters (e.g. similarity index, accuracy index and concordance index, among others) (Mombelli et al., 2016). If the assessed compound is present in the dataset used to build the model, the experimental value is provided by the model with the associated notification "Experimental value" and the ADI is not taken into account. Otherwise, predictions are categorized into good, moderate and low reliability based on the calculated ADI value. The consensus score (CS), also ranging from 0 to 1, applies to the consensus model available for genotoxicity endpoint and combines the predictions of the different models included and their associated level of reliability (Mombelli et al., 2022).

2.1.3. Implementation of the information sources and (Q)SAR models in the SILIFOOD tool

After selecting the different information sources and (Q)SAR models, the structure of the SILIFOOD tool was designed (representation in Fig. 1) and implemented into a software application freely available within the VEGAHUB platform (www.vegahub.eu).

2.2. Case studies

2.2.1. Compound selection

The applicability of the SILIFOOD tool was investigated by performing case studies with non-evaluated substances migrating from FCM identified based on a literature search. Notifications in the RASFF related to the migration of compounds from FCM were also considered. The case study compounds were selected based on the following criteria: (i) the evaluation status for use in FCM, (ii) the existence of migration or extraction data indicating possible exposure, and (iii) the amount of available toxicological data. To assess the evaluation status, the inclusion of the compound in the positive list of substances authorized for their use in plastic FCM (Annex I of Regulation (EU) No. 10/2011) was verified. Only substances not evaluated for their use in FCM (non-plastic FCM and NIAS) were considered for the selection. The existence of migration/exposure data was investigated by consulting the Database on Migrating and Extractable Food Contact Chemicals (FCCmigex), an interactive database that consists of the inventory of scientific evidence of FCCs that have been measured in migrates and extracts of FCM (Geueke et al., 2022). Finally, only compounds for which a considerable amount of toxicological information was present in the data sources included in SILIFOOD were selected. This allowed to evaluate whether these data could also be collected with the tool.

2.2.2. Application of the SILIFOOD tool for the selected compounds

For each of the selected compounds, the CAS number and the SMILES were retrieved on PubChem and used to run the SILIFOOD tool. The generated PDF reports, either based on the CAS number or the SMILES, were compared to verify the consistency when using different chemical identifiers. In addition, all data sources were consulted manually to check the ability of the SILIFOOD tool to automatically retrieve for a specific case study compound all the data present in the data sources included in the tool. Afterwards, the toxicological information and predictions provided by the tool were analyzed in more detail.

2.2.3. Use of data collected with SILIFOOD to support the risk assessment of the case study compounds

The data collected by the SILIFOOD tool were combined with exposure data, collected manually, to perform a (preliminary) risk assessment according to the RACE methodology of EFSA (Fürst et al., 2019). For FCM, the methodology starts with a pre-decision tree to determine if the migrating compound is covered by European or national legislation, and if so, whether the legal limits are exceeded. The regulatory information needed in the pre-decision tree was collected automatically with the SILIFOOD tool from the *in-house* FCM database of Sciensano. If the migration of the chemical did not exceed the legal limit, it was considered that no safety concern was expected for human health and that there was no need for an additional evaluation. However, if the limit was exceeded or no legislation was in place for the FCM compound, the pre-decision tree was followed by the risk evaluation decision tree.

In the first part of the risk evaluation, it was checked whether the substance is a genotoxic carcinogen, as this will drive the further risk evaluation process. The genotoxic and carcinogenic potential of the compound was estimated based on the information retrieved from different sources included in the SILIFOOD tool and/or the prediction outcomes of the VEGA models.

If the compound was not (suspected to be) a genotoxic carcinogen, the next step of the decision tree consisted of finding an HBGV (acute and/or chronic) such as an Acute Reference Dose (ARfD), an Acceptable or Tolerable Daily, Weekly or Monthly Intake (ADI/TDI/TWI/TMI). When a HBGV was available, the FCM compound exposure was compared to this HBGV. When no HBGV was available for the target compound, information on RPs (e.g. NOAEL) was collected and the margin of safety (MoS) (ratio between the RP value and the estimated exposure) was calculated. When available, HBGVs and/or RPs were collected by the SILIFOOD tool from the Council of Europe Resolutions and/or from EFSA data/opinions present in the OpenFoodTox database. In case no HBGVs or RPs were available, the NOAEL prediction generated by the VEGA model could be used under the condition that the level of reliability was good and bearing in mind that an additional uncertainty factor might be needed. If no information was available to collect/derive a RP value, the TTC approach was applied based on the information on the Cramer classes provided by SILIFOOD. However, before applying the TTC approach, the applicability criteria and exclusion categories must be considered by the user of the SILIFOOD tool, as described by EFSA (EFSA Scientific Committee et al., 2019).

If the substance was identified as a genotoxic carcinogen, an RP was

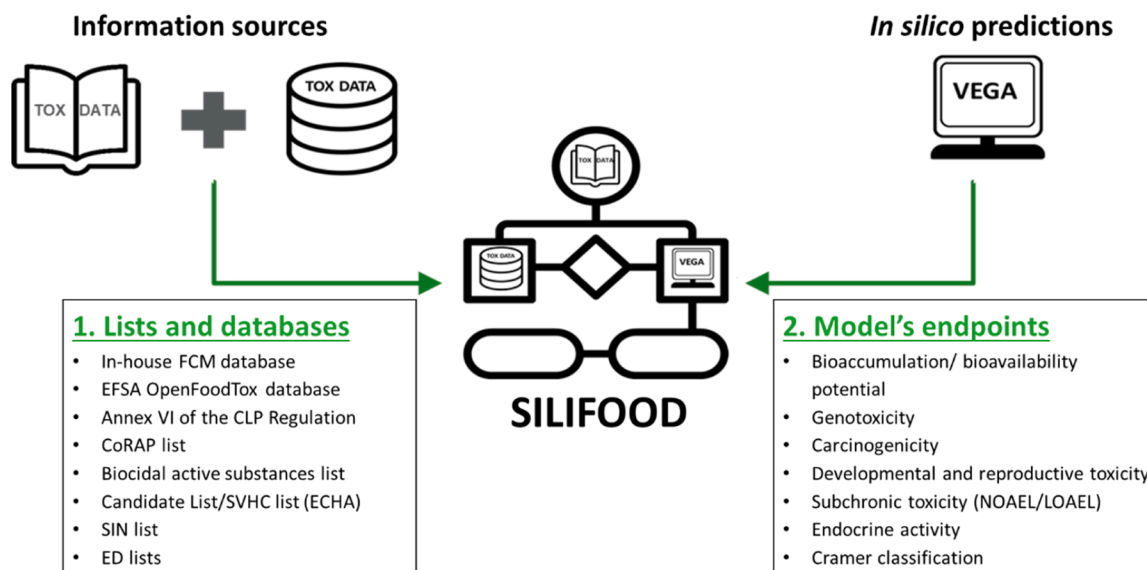


Fig. 1. Implementation of the information sources and in silico predictions in the SILIFOOD tool.

identified to apply the adequate Margin of Exposure (MoE) approach, which, as the MoS, involves comparing the available RP to the estimated exposure. If no adequate RP was available, the TTC approach was applied using the specific value of $0.0025 \mu\text{g kg}^{-1} \text{bw day}^{-1}$ associated with genotoxic and carcinogenic compounds (with the exception of exclusion criteria).

Besides the RACE-based risk assessment, information collected with

the SILIFOOD tool was also used to assess the ED potential of the compounds. ED-related information was obtained by checking the presence of the compound of interest in the lists with identified or suspected ED compounds. Secondly, existing data related to endocrine activity were collected by consulting the reports present on the EFSA OpenFoodTox database or provided in the frame of inclusion on the CoRAP list, if available. Finally, the predictions from the VEGA models related to

Table 1

Overview of the selected lists and databases and the information extracted from these data sources. SML: Specific Migration Limit, HBGV: Health Based Guidance Value, CMR: Carcinogenic, Mutagenic, toxic for Reproduction, ED: Endocrine Disruptor.

List and database	Description	Information of interest present in the data source				
		Evaluation status	Legal limit (SML) - Restrictions & specifications	HBGV or reference value	CMR information	ED information
Sources included in the in-house FCM database						
Annex I Regulation (EU) 10/2011	List of substances authorized for plastics FCM (Commission Regulation (EU) 2011)	X	X			
Synoptic Document	Inventory of monomers and additives used in plastics and coatings (European Commission, 2005)	X	X	X		
Swiss Ordinance Annex 10	Lists of permitted substances for the manufacture of packaging inks (previously Annex 6 of the Swiss Ordinance) (FDHA/FSVO, 2020)	X	X		X	
Council of Europe (CoE) Resolutions (Paper & Board, Coatings, Cork, Silicones)	Policy statements containing inventories for substances used in (the manufacturing of) the particular type of material in contact with food, listed based on notifications of the industry and/or local and national authorities. (Resolutions & Policy Statements 2023)	X	X	X		
EFSA ESCO reports (Paper & Board, Cork & Wood, Silicones, Rubber, Colorants)	Collection of substances used in non-plastic FCM and present in positives lists of European countries. (European Food Safety Authority, 2011)	X	X			
Additional publicly available lists and databases						
EFSA OpenFoodTox database	Database providing an overview of the main toxicological data present in the scientific EFSA opinions (Substance characterization, genotoxicity study, reference value(s) and reference point(s) together with the link to the reference opinion) (Chemical Hazards Database (OpenFoodTox) EFSA, 2021)	X	X	X	X	X
Annex VI of the CLP Regulation	Table of harmonized entries of the CLP Regulation (EC) No 1272/2008. (Table of Harmonised Entries in Annex VI to CLP - ECHA 2023)	X			X	X
CoRAP list	The Community Rolling Plan (CoRAP) list contains substances evaluated or that will be evaluate by a Member State under the REACH Regulation (EC) No 1907/2006. (Substance Evaluation - CoRAP - ECHA 2023)	X			X	X
Biocidal active substances list	Provide information on the evaluation status of substances concerned by the Biocidal Product Regulation (EU) No 528/2012. (List of Approved Active Substances - ECHA 2023)	X				
Candidate list /SVHC list (ECHA)	List the Substances identified to be of Very High Concern (SVHC) by ECHA, Member states or the European Commission based on the criteria established in article 57 of Regulation (EC) No. 1907/2006. (Candidate List of Substances of Very High Concern for Authorisation - ECHA, 2023)				X	X
SIN List	The Substitute it Now (SIN) list includes substances identified by the International Chemical Secretariat (ChemSec, Gothenburg, Sweden) to be of Very High Concern (SVHC) based on the criteria established in article 57 of Regulation (EC) No. 1907/2006. (SIN List 2022)				X	X
ED Lists	Lists of substances identified or suspected to be EDs by the European or National authorities: - List I: Substances identified as endocrine disruptors at EU level. - List II: Substances under evaluation for endocrine disruption under an EU legislation. - List III: Substances considered, by the evaluating National Authority, to have endocrine disrupting properties. (Endocrine Disruptor List, 2023)					X

endocrine activity were also considered, as they provide, in some cases, relevant insights into the ED potential.

3. Results and discussion

3.1. Development of the SILIFOOD tool

3.1.1. Selection and curation of lists and databases

Several lists and databases were selected based on different criteria such as the toxicological data content, the availability, and the possibility of including these information sources in an automated tool. Three categories of lists and databases were considered: (i) data sources specifically related to FCM, (ii) data sources related to substances evaluation in a non-FCM context and (iii) lists of substances with toxicological concern (e.g. suspected or confirmed ED). The selected databases/sources and the relevant information for FCM risk assessment are summarized in Table 1. To facilitate the inclusion of information from extensive databases, such as the EFSA and ECHA databases, in the SILIFOOD tool, specific strategies were adopted. As a result, it is essential to note that the EFSA OpenFoodTox database was included in a manner that allows the SILIFOOD application to provide references for the EFSA opinions available in the EFSA OpenFoodTox database, but does not automatically extract the toxicological data. Moreover, the inclusion of the entire ECHA dataset was not possible for several reasons. First, the amount of data the manufacturer provides is vast, and most of this information has not (yet) been verified by competent authorities. Second, the inclusion of the very large list of registered substances was not possible within the limited timeframe of the project. Instead, ECHA lists of evaluated substances were included.

For each of the selected lists and databases, KNIME workflows were applied to generate and/or curate the SMILES of the different substances. The OpenFoodTox database already contained the Original SMILES; consequently, only curation was applied to obtain the VEGA SMILES for this database. For the other lists and databases, the semi-automated workflow described by Gadaleta et al. (2018) was applied to retrieve and curate the SMILES. Since the latter requires the CAS number in addition to the substance name, this workflow cannot retrieve SMILES for substances without a CAS number. In addition, this semi-automated KNIME workflow automatically rejects substances for which no SMILES are available and, therefore, for which VEGA models cannot be applied (*i.e.*, mixtures, polymers, degradation products, reaction masses/products, inorganic and organometallic compounds, chemical substances of unknown or variable composition, and biological materials). Overall, the average percentage of substances without SMILES among all the lists and databases is around 40 %, including 10 % of substances without a CAS number and 30 % of rejected substances. The semi-automated KNIME workflow also strips and neutralizes salts, and does not keep any information about stereoisomers. For these reasons and to provide reliable and complete data, the scope of the tool is limited to single organic substances.

3.1.2. In silico methods to predict bioavailability/bioaccumulation potential

A sine qua non condition for a substance ingested orally by humans to demonstrate/exhibit a systemic (toxicological) effect is that the substance is absorbed and enters the blood circulation. Consequently, the substance must be able to cross the biological barriers of the gastrointestinal tract. Some chemical structure properties can be used to predict/estimate the bioavailability/bioaccumulation potential. Using the criteria described by Van Bossuyt et al. (2016) based on the “Lipinski rule of 5”, molecular descriptors were integrated into the SILIFOOD tool to calculate the molecular weight, number of hydrogen bond donors and acceptors, polar surface area, and the number of rotating bonds. In addition, a LogP model, *i.e.*, the Meylan/Kowwin model, was also included (Table 2). The logP value can also be used to evaluate the bioavailability of the substance and indicate its potential for bioaccumulation.

Table 2

Overview of the (Q)SAR models selected from VEGA. NOAEL: No Observed Adverse Effect Level; TTC: Threshold of Toxicological Concern.

Toxicological data/ Endpoint	Models selected from VEGA (version 1.2.0)
Data to demonstrate the absence of accumulation potential in man	• LogP model (Meylan/Kowwin) (version 1.1.5)
Genotoxicity	• Mutagenicity (Ames test) CONSENSUS model (version 1.0.4)
• Gene mutations	• Chromosomal aberration model (CORAL) (version 1.0.1)
• Structural and numerical chromosome aberration	• <i>In vitro</i> Micronucleus activity (IRFMN/Vermeer) (version 1.0.1)
	• <i>In vivo</i> Micronucleus activity (IRFMN) (version 1.0.2)
Data from long term toxicity/carcinogenicity studies	• Carcinogenicity model (CAESAR) (version 2.1.10)
	• Carcinogenicity model (ISS) (version 1.0.3)
	• Carcinogenicity model (IRFMN/ISSCAN—CGX) (version 1.0.1)
	• Carcinogenicity model (IRFMN/Antares) (version 1.0.1)
	• Carcinogenicity oral classification model (IRFMN) (version 1.0.1)
Data on reproductive and developmental toxicity	• Developmental Toxicity model (CAESAR) (version 2.1.8)
	• Developmental/Reproductive Toxicity library (PG) (version 1.1.2)
Subchronic oral toxicity data (90-day study)	• NOAEL (IRFMN/CORAL) (version 1.0.1)
Endocrine disruption	• Estrogen Receptor Relative binding Affinity Model (IRFMN) (version 1.0.2)
	• Estrogen receptor-mediated effect (IRFMN/CERAPP) (version 1.0.1)
	• Androgen receptor-mediated effect (IRFMN/COMPARA) (version 1.0.1)
	• Thyroid receptor alpha effect (NRME) (version 1.0.1)
	• Thyroid receptor beta effect (NRME) (version 1.0.1)
Information to support the TTC approach	• Cramer classification model (Cramer rules, ToxTree) (version 1.0.1)

3.1.3. Selection of (Q)SAR models for hazard predictions

For each endpoint of interest (*i.e.*, genotoxicity, carcinogenicity, developmental and reproductive toxicity, and sub-chronic toxicity (NOAEL determination), endocrine activity), at least one model (classification model or quantitative model) was identified in VEGA (Table 2).

In addition, a model for the Cramer class was also included, as this information can support the application of the TTC approach in case insufficient data are available to perform the risk assessment. More details of the selected models can be found on the VEGA website, where they are described in their associated reference files (<https://www.vegahub.eu/portfolio-item/vega-qsar-models-qrmf/>).

3.1.4. Implementation of information sources and (Q)SAR models in SILIFOOD

The databases/lists and (Q)SAR models were integrated in a workflow consisting of 7 steps:

- Step 1. Molecule input/ Identification of the substance
- Step 2. Data collection: Evaluation data in the context of FCM use
- Step 3. Data collection: Evaluation data in a non-FCM context

- EFSA database
- ECHA lists of evaluated compounds

- Step 4. Presence on a specific list of toxicological concern
- Step 5. Bioavailability/bioaccumulation potential assessment

Step 6. Hazard identification by (Q)SAR predictions

- Genotoxicity
- Carcinogenicity
- Reproductive and developmental toxicity
- NOAEL (subchronic oral toxicity data: 90-day study)
- Endocrine activity

Step 7. Cramer classification prediction to enable the application of the Threshold of Toxicological Concern (TTC) approach

This workflow was implemented and automated in a publicly accessible software platform (<https://www.vegahub.eu/portfolio-item/silifood/>). The final result is a stand-alone application, namely the SILIFOOD application, developed in Java with a straightforward and user-friendly interface to collect data and QSAR predictions (Fig. 2). The SILIFOOD tool has been developed specifically to support the assessment of non-evaluated FCM substances and focuses on single organic substances. The application can be run using the CAS number, the SMILES, or the FCM number as a chemical identifier and generates a PDF report with the collected data in a few seconds. Although the tool has been developed for single organic substances, and salts are considered outside its applicability domain, it is possible to generate a SILIFOOD report for salts by introducing both the CAS number and the SMILES. This is due to the fact that the CAS number is required to collect literature data while the SMILES are used by the VEGA model to provide hazard predictions. Introducing only one identifier for salts could lead to an incomplete report. Moreover, it is not possible with this tool version to collect data about salts using only the FCM number. Detailed information on how to run the application can be found in the user manual accessible online (<https://www.vegahub.eu/wp/wp-content/uploads/2023/11/SILIFOOD-User-Manual.pdf>).

3.2. Case studies

3.2.1. Compounds selection and application of the SILIFOOD tool

Based on the criteria mentioned in Section 2.2.1, the following substances were selected: perfluorooctane sulfonic acid (PFOS), diisobutyl phthalate (DiBP), 2,4-di-tert-butylphenol (2,4-DtBP) and 2-isopropylthioxanthone (ITX). A detailed overview of the chemical identifiers of the compounds and the reasons for selection are provided in Table 3. For these four substances, the reports generated by the SILIFOOD tool using either their CAS number or their SMILES as chemical input were identical, i.e. the same information was obtained in the reports regardless of the chemical input/identifier used. Additionally, the information collected by the SILIFOOD tool was comparable to the data collected manually from the data sources included in the workflow. These observations demonstrate the tool's ability to consistently provide all data available in the data sources included in the workflow. A summary of the data collected by the SILIFOOD application for each of the selected substances is provided in Table 4.

3.2.2. Risk assessment

A risk assessment was performed for each selected compound (i.e. DiBP, 2,4-DtBP, ITX, PFOS) using the data collected by the SILIFOOD application following the EFSA RACE methodology and the exposure scenario described in Table 3. Regarding the data collected by the SILIFOOD application, the priority is given to the existing toxicological data collected from identified/included data sources. Then, in cases of absence or incomplete hazard assessment information, *in silico* predictions can be considered, giving more weight to the experimental results provided by VEGA and to the predictions with the highest reliability, where reliability is based on the applicability domain index determined for each prediction of VEGA models. Moreover, results should be interpreted within a weight-of-evidence process, and critical scientific/expert judgment is essential, playing a key role in decision-

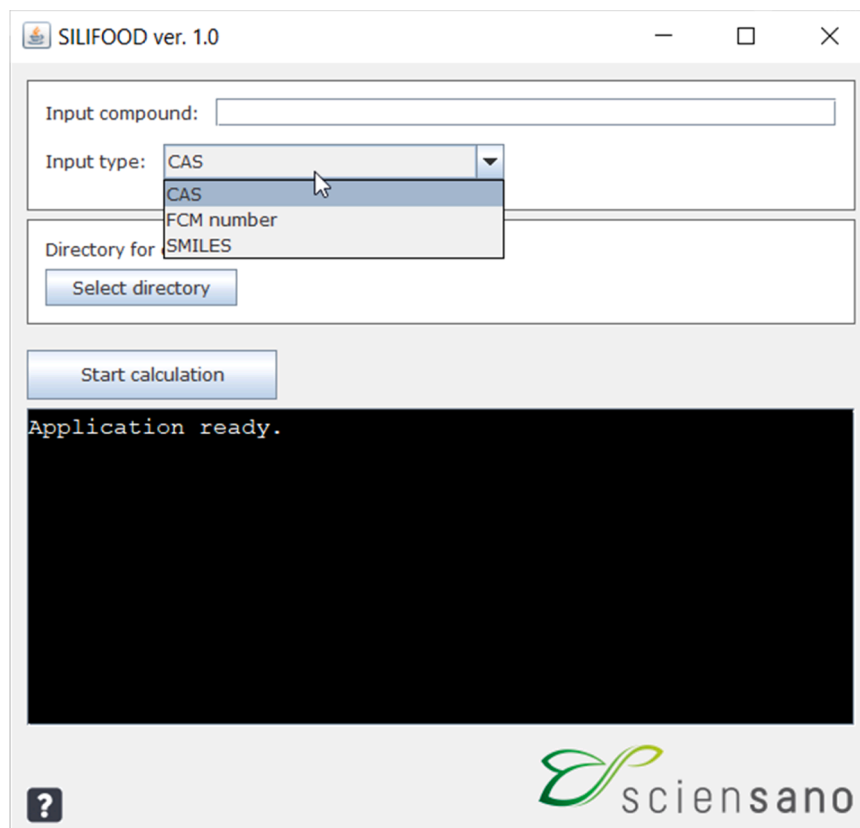


Fig. 2. Interface of the SILIFOOD application.

Table 3

Selected compounds for the case studies and associated CAS number and SMILES. CLP: Classification, Labelling and Packaging; ED: Endocrine Disruptor; EFSA: European Food Safety Authority; FCC: Food Contact Chemical, FCM: Food Contact Material; NIAS: Non-Intentionally Added Substance; SVHC: Substance of Very High Concern; PFAS: perfluoroalkyl and polyfluoroalkyl substances; POP: Persistent Organic Pollutants; REACH: Registration, Evaluation, Authorization and Restriction of Chemicals; Repr.: Reproductive toxicity; SMILES: Simplified Molecular Input Line Entry System.

Compound's name	CAS No	SMILES	Reason for selection	Exposure data
Perfluorooctane sulfonic acid (PFOS)	1763–23–1	<chem>C(C(C(C(C(F)(F)S(=O)(=O)O)(F)F)(F)F)(F)F)(C(C(C(F)F)F)F)(F)F)F</chem>	- Belongs to PFAS which are synthetic chemicals widely used in FCM (paper, cardboard, cookware,...). - Migration to food is demonstrated (Ramírez Carnero et al., 2021) and 16 data entries are present in the FCCmigex database (12 references). - Not included in Annex I of Regulation (EU) No. 10/2011. - PFAS are classified as persistent, bioaccumulative and toxic substances associated with cancer, developmental toxicity and immunotoxicity. - In 2009, PFOS was restricted by the Stockholm Convention and its use is now prohibited due to its inclusion in Annex I of the POP Regulation. PFOS should only be present in FCM as NIAS.	A maximum level of PFOS of 92.48 ng dm ⁻² was extracted from a fried chicken box (Poothong, Bootnanon and Boontanon, 2012). This concentration was divided by two as it is assumed that only 50 % of extracted compound migrates (Bokkers et al., 2019). Based on this concentration and the estimation that a person of 60 kg eats 1 kg of food packed in 6 dm ² per day, the exposure was estimated to be 4.6 ng kg ⁻¹ bw day ⁻¹ .
Diisobutyl phthalate (DiBP)	84–69–5	<chem>CC(C)COC(=O)C1=CC=CC=C1C(=O)OCC(C)C</chem>	- Migration of DiBP from plastics and other types of FCM reported: 211 entries and 100 references in the FCCmigex database (Consulted in 2023). - Not included in Annex I of Regulation (EU) No. 10/2011. - Identified as NIAS (Kato and Conte-Junior, 2021). - Classified as toxic to reproduction (Repr.1B) by the CLP Regulation, as SVHC under REACH and identified as ED at EU level (list I of on the edlists.org website).	Migration from a polycarbonate replacement baby bottle was previously reported (Onghena et al., 2016). The worst-case exposure scenario was estimated to be 3.5 µg kg ⁻¹ bw day ⁻¹ assuming a drinking volume of 233 mL kg ⁻¹ (infant of 3 kg consuming 100 mL 6–7 times a day) (Mertens et al., 2016).
2,4-di-tert-butylphenol (2,4-DtBP)	96–76–4	<chem>CC(C)(C)C1=CC(=C(C=C1)O)C(C)(C)C</chem>	- Degradation product of the antioxidants Irganox 168, Irganox 1076 and Irganox 110 used in FCM manufacturing. 64 references in the FCCmigex database and 135 database entries from 1995 to 2021). - Not included in Annex I of Regulation (EU) No. 10/2011 - Identified as NIAS in different FCM (Kato and Conte-Junior, 2021). - No toxicological data found in previously performed risk assessment (Mertens et al., 2016).	Migration was reported from 5 out of 24 polycarbonate replacement baby bottles analysed (Onghena et al., 2016). In the worst-case exposure scenario, the maximum daily exposure of 27.5 µg kg ⁻¹ bw day ⁻¹ was calculated assuming a drinking volume of 233 mL kg ⁻¹ (infant of 3 kg consuming 100 mL 6–7 times a day) (Mertens et al., 2016).
2-isopropylthioxanthone (ITX)	5495–84–1	<chem>CC(C)C1=CC2=C(C=C1)SC3=CC=CC=C3C2=O</chem>	- Photo-initiator frequently used as a catalyst in the manufacturing of UV-cured inks (Van Bos-suyt et al., 2016). - Subject of food contamination crisis in 2005: presence of ITX in liquid milk for babies packaged in printed cardboard. 50 data entries found in the FCCmigex database from 26 references demonstrating its migration from different types of FCM. - Not included in Annex I of Regulation (EU) No. 10/2011. - Lack of toxicity data hampering the health risk assessment of ITX.	Based on the concentration value of 250 µg L ⁻¹ of ITX in contaminated milk, the EFSA Scientific Panel calculated a potential dietary exposure in infants (fed exclusively with liquid infant formulae packaged in cardboard printed with UV-cured inks) at the 95th percentile of consumption equal to 43 µg kg ⁻¹ bw day ⁻¹ (Authority (EFSA) 2005).

making.

The toxicological data and risk assessments were then compared with (previous) risk assessments performed with data collected manually (Supplementary Tables 1–4).

The information collected from the FCM database confirmed the absence of a legal limit for plastic FCM for the four compounds, as none of the compounds was included in Annex I of the EU plastic Regulation. Then, toxicological data such as genotoxicity and/or carcinogenicity results, HBGV(s), or RP(s) were considered. Some of this toxicological information was directly displayed in the SILIFOOD reports while, in other cases (e.g., for the OpenFoodTox database, the CoRAP list, or the biocidal active substances list from ECHA), the links to associated documents containing additional data were provided.

3.2.2.1. PFOS. The SILIFOOD report indicated that PFOS is classified as carcinogenic and toxic for reproduction under the CLP Regulation. In an EFSA opinion from 2018, for which the link was provided in the SILIFOOD report, PFOS was reported to be a tumour promotor. EFSA also concluded that there was no evidence for genotoxicity and derived a tolerable weekly intake (TWI) of 13 ng kg⁻¹ bw week⁻¹ (EFSA, 2018). Based on this HBGV, the estimated PFOS exposure of 32.2 ng kg⁻¹ bw week⁻¹ (4.6 ng kg⁻¹ bw day⁻¹ calculated based on the migration study of Poothong et al. (2012) (Table 3)) may pose a health risk for the consumers. The risk assessment performed with the data collected manually (Supplementary Table 1) resulted in the same conclusion. However, during the manual data collection, a more recent EFSA opinion mentioning a TWI of 4.4 ng kg⁻¹ bw week⁻¹ for the sum of four

Table 4

Overview of the data collected for the selected case study compounds with the SILIFOOD application, including the presence/absence of the compound in the different selected data sources and the predictions of the selected VEGA models. CLP: Classification, Labelling and Packaging; CMR: Carcinogenic, Mutagenic and toxic for Reproduction; CoE: Council of Europe; CoRAP: Community Rolling Action Plan; DB: Database; ECHA: European Chemicals Agency; ED: Endocrine disruptor; EFSA: European Food Safety Authority; ESCO: EFSA Scientific Cooperation; FCM: Food Contact Material; NOAEL: No Observed Adverse Effect Level, SIN: Substitute It Now; SVHC: Substance of Very High Concern.

Information sources	Literature data												<i>In silico</i> model predictions									
Chemical identification	FCM DB					EFSA DB	ECHA DB			Specific lists		VEGA models										
Compound	Annex I	Synoptic document 2005	Swiss Ordinance	ESCO report(s)	CoE Resolution(s)	EFSA OpenFoodTox database	CLP harmonized (CMR)	CoRAP list	Biocidal Active Substances list	SVHC	SIN list	ED lists	Bioavailability (estimated)	Mutagenicity (Consensus)	Chromosomal aberration	<i>In vitro</i> micronucleus	<i>In vivo</i> micronucleus	Carcinogenicity	Developmental/ Reproductive Toxicity	Endocrine activity	NOAEL (mg kg ⁻¹ bw)	Cramer Classification
Perfluorooctane sulfonic acid (PFOS)	-	-	-	-	-	Yes	Yes	-	-	-	Yes	-	-	(-)	(-)	(-)	+	-/(+)	+	-	(2)	III
Diisobutyl phthalate (DiBP)	-	Yes	Yes	Yes	Yes	Yes	Yes	-	-	Yes	Yes	List I	+	-	-	-	-	-/+	(-)/-	+	38	I
2,4-di-tert-butylphenol (2,4-DtBP)	-	-	Yes	-	Yes	-	-	Yes	-	-	-	List II	+	-	-	+	-	-	(-)/+	+	7	I
2-Isopropylthioxanthone (ITX)	-	-	Yes	-	Yes	Yes	-	-	-	-	-	-	+	(+)	+	(+)	-	-	+	(-)/+	(2)	III

-	Overall negative model output(s)
-/+	Inconclusive model output(s)
+	Overall positive model output(s)
()	Prediction with low reliability

perfluoroalkyl substances (PFOA, PFNA, PFHxS and PFOS) was found (EFSA CONTAM Panel et al., 2020). This EFSA opinion was not yet included in the EFSA OpenFoodTox database at the time of the case study which explains why it could not be retrieved by the SILIFOOD tool. The current risk assessment did not consider PFOS originating from other exposure sources and/or co-exposure with PFOA, PFNA and PFHxS, as only extraction data of PFOS from food packaging were provided in the publication of Poothong et al. (2012). PFOS was not present on any of the ED lists. In addition, all the VEGA predictions for endocrine activity were negative. These data are in agreement with the EFSA Opinion of 2020 wherein no association between PFOS and endocrine effects was found.

3.2.2.2. DiBP and 2,4-DtBP. No harmonized CLP classification for carcinogenicity or mutagenicity was found for either DiBP or 2,4-DtBP. However, the SILIFOOD tool indicated the presence of DiBP in the harmonized classification list as reprotoxic (Repr.1B) and on the Candidate list (ECHA) as toxic for reproduction and ED properties. Moreover, according to the SILIFOOD tool, 2,4-DtBP is present on the CoRAP list as suspected carcinogenic, mutagenic, or toxic for reproduction (CMR) and potential ED. No HBGV or RPs were collected with the SILIFOOD tool for DiBP and 2,4-DtBP, and consequently, there was insufficient information to perform a risk assessment. In order to (partly) fill the data gaps, the use of the VEGA (Q)SAR model predictions provided by the SILIFOOD tool to perform a preliminary risk assessment was explored. However, when using these predictions, it is important to consider the reliability level associated with the predictions. Overall, the VEGA models predicted both DiBP and 2,4-DtBP as non-mutagenic. The

mutagenicity model (consensus model) predicted DiBP and 2,4-DtBP as non-mutagenic with a high consensus score (1 and 0.9, respectively) and the three other genotoxicity models predicted DiBP as inactive/non-genotoxic with moderate (for the *in vitro* micronucleus activity) and good (for the chromosomal aberration and the *in vivo* micronucleus activity) reliability. Although the *in vitro* micronucleus activity was predicted positive for 2,4-DtBP (with good reliability), the *in vivo* micronucleus activity model and the chromosomal aberration models predicted the compound non-genotoxic (good reliability).

Next, the predictions of the subchronic oral toxicity model were considered. For DiBP, a NOAEL of 38 mg kg⁻¹ bw day⁻¹ was predicted with good reliability whereas for 2,4-DtBP, an experimental NOAEL value of 6.67 mg kg⁻¹ bw day⁻¹ was provided. Based on these NOAEL values and the exposure data (Table 3), the MoS were calculated. For DiBP, the MoS was equal to 10 877. This value is thus >5x larger than the overall default uncertainty factor (UF) of 200–300 (10 for inter-species variability x 10 for intra-human variability x 2 or 3 for extrapolation from subchronic to chronic duration). For 2,4-DtBP, the resulting MoS would be 242.5, which is still larger than 200 when a factor of 2 for extrapolation is used, according to the EFSA Guidance on selected default values (EFSA Scientific Committee, 2012). Although these results indicate that the risks associated with the exposure to DiBP and 2,4-DtBP are low, it should be emphasized that there is a lot of uncertainty related to the use of predictions of NOAEL (Manganelli et al., 2023). Moreover, when considering the margin of 300 (as recommended by Fürst et al.), the exposure of 2,4-DtBP becomes of concern.

In contrast, the risk assessments for DiBP and 2,4-DtBP based on data collected manually (Supplementary Tables 2 and 3, respectively) were

slightly different since at least one HBGV or RP could be retrieved in the ECHA database. Indeed, when a chemical is registered under REACH, toxicological data provided by the manufacturer can be found on the ECHA website as well as in the opinion from the Risk Assessment Committee (RAC) of ECHA, if available. As no link to these toxicological information from the ECHA database was included in our workflow, this information was not automatically retrieved by the tool and needed to be collected manually.

For 2,4-DtBP, an oral DNEL value of $3.75 \text{ mg kg}^{-1} \text{ bw day}^{-1}$ and a NOAEL value of $150 \text{ mg kg}^{-1} \text{ bw day}^{-1}$ were found in the ECHA database (for the general population based on repeated dose toxicity via oral route). When using these values for the risk assessment, the conclusion is a low concern for adverse health effects (details of the calculation can be found in Supplementary Table 3). Interestingly, the experimental NOAEL value provided by the VEGA model is lower and therefore more protective than the NOAEL found on the ECHA database.

Concerning the ED properties, the SILIFOOD tool reported that both compounds are present on the ED lists and at least one other inventory list with concern about ED effects. DiBP is present on the ED list I, including the substances identified as ED at the EU level, and on the candidate list of ECHA as ED to humans. 2,4-DtBP is present on the ED list II, listing the substances under evaluation for ED under EU legislation and on the CoRAP list as suspected ED. These data were in line with the ED-related information collected manually.

A previous risk assessment performed for DiBP with data collected manually resulted in a similar conclusion as the risk assessment performed with data collected by SILIFOOD (see Supplementary Table 2) except that DiBP is now identified as ED while it was, at that time, indicated as suspected ED (Category 2 of EU priority list) (Mertens et al., 2016).

3.2.2.3. ITX. For ITX, two EFSA opinions (Authority (EFSA), European Food Safety, 2005, Authority (EFSA), European Food Safety, 2007) were found by the SILIFOOD tool. Within each of these EFSA opinions, the genotoxic potential of ITX was assessed with similar conclusions, *i.e.* negative based on the outcome of *in vivo* studies, although the *in vitro* studies provided contradictory results. No carcinogenicity data were available in any of the two EFSA opinions, and only the results of one repeated dose 28-day oral toxicity study in rodents were provided in the EFSA opinion from 2007 (Authority (EFSA), 2007). However, due to the short duration of the study, EFSA did not derive an HBGV or RP based on this study. No other reference value could be collected by the SILIFOOD tool and the NOAEL provided by the VEGA model was not taken into account due to the low reliability of the prediction. In this case and in the absence of exclusion criteria, the TTC approach is recommended by the EFSA RACE methodology. The ToxTree model included in the SILIFOOD tool provides the Cramer class in the SILIFOOD report and indicates ITX as belonging to the Cramer Class III associated with a threshold of $1.5 \mu\text{g kg}^{-1} \text{ bw day}^{-1}$. In this case, the estimated exposure of ITX ($43 \mu\text{g kg}^{-1} \text{ bw day}^{-1}$, see Table 3 for the detailed calculation) is higher than the threshold. Therefore, based on the data collected by the SILIFOOD application and the TTC approach, this level of ITX in FCM may pose a health risk for the consumer. This is in contrast with the outcome of the risk assessment performed with data collected manually. Indeed, the exposure is below the DNEL of $0.1 \text{ mg kg}^{-1} \text{ bw day}^{-1}$ found after manual consultation of the ECHA website, indicating that the migration of ITX is of low concern for consumer health (Supplementary Table 4). However, it is important to highlight that ITX is currently under assessment for suspected mutagenic effects and suspected ED activity and more data are requested by ECHA as attested by the Assessment for Regulatory Needs (ARN) published in 2022 (ECHA, 2022). This source of information could not be integrated in the SILIFOOD tool since no inventory of these documents was available.

Regarding ED activity, ITX was not present on any list of potential or suspected ED, and only one model prediction was “active”. Depending

on the outcome of the ECHA ARN recommendations, this might change in the future. As the data sources in SILIFOOD will be updated on a regular basis, these possible modifications in the compound's ED status will be reflected in the generated SILIFOOD reports.

3.2.3. Strengths, limitations, and future perspectives

- Overall, the case studies demonstrated the high efficiency and speed of the SILIFOOD tool in collecting toxicological data compared to manual data collection, as information from different data sources was correctly obtained in only a few seconds. Moreover, the inclusion of *in silico* models in the tool offers the possibility to fill data gaps. Although an additional manual search can be useful for finding information not included in SILIFOOD, the case studies showed that the risk assessments performed with the SILIFOOD tool were, overall, equal to or more protective than those performed with manually collected data. A practical application of the tool was also presented in the work of Di Mario et al. (2023). It should, however, be noted that the tool cannot currently be run simultaneously for multiple chemicals (as is the case with other *in silico* tools or platforms, such as VEGA). Simultaneous analysis of a ‘batch’ of compounds could be interesting for assessing a group of chemicals sharing an HBGV (*e.g.*, TWI provided for four perfluoroalkyl substances), mixtures of several single organic chemicals, read-across, chemical comparison, or prioritization.
- The data collected and provided by SILIFOOD depends on the tool's integrated data. When data is not inventoried in a manner to allow its easy integration in SILIFOOD (as is the case for the assessment for regulatory needs available on the ECHA website) or if the lists or databases used as sources are incomplete (*e.g.*, the EFSA report that was not yet integrated into the EFSA OpenFoodTox database), SILIFOOD will logically be unable to find them. This implies that the data sources (lists and databases) should be regularly updated to ensure the SILIFOOD tool provides the most up-to-date data possible. For this reason, regular updates of the SILIFOOD tool are foreseen, and the last update of each source is well documented in the SILIFOOD report. The integration of additional lists or databases in SILIFOOD could be the object of future development to collect additional regulatory and toxicological data automatically.
- The inclusion of (Q)SAR models in SILIFOOD is a good way to collect hazard information/insight for chemicals with toxicological data gaps. Including the ToxTree model (Cramer rules) to classify the chemical into its appropriate Cramer class allows the risk assessment of data-poor chemicals (as demonstrated by the case study with ITX). Another possibility to collect hazard data or to perform a risk assessment for data-poor chemicals is to apply read-across with (structurally) similar compounds/chemicals. At this moment, the tool was designed to collect data for any individual organic compound for which the CAS number and/or SMILES are known. Nevertheless, the tool cannot identify chemical similarities and automatically provide data for similar chemicals. Adding this functionality to SILIFOOD could also offer a perspective for further development of the tool.
- Finally, the SILIFOOD tool was designed to collect hazard data, but this data has to be combined manually with exposure data to perform chemical risk assessment. The VERMEER FCM (Van Overmeire et al., 2024) application includes, in addition to regulatory and hazard modules, also a module to predict migration of chemicals from FCM. Consequently, the VERMEER FCM and SILIFOOD tools provide complementary information and could be used in combination. However, it should be noted that VERMEER FCM was specifically developed for plastic FCM, while the SILIFOOD tool has a larger scope, including non-plastic FCM chemicals.

4. Conclusion

The SILIFOOD tool, which is freely available, was developed to collect regulatory and toxicological data to support the risk assessment of non-evaluated FCM substances. The methodology proposed in the RACE tool of EFSA was used as a starting point for the development of the tool. Sources and databases containing relevant existing regulatory and toxicological data were included in the tool with a focus on single organic substances. Including (Q)SAR models from VEGA in the SILIFOOD tool allows the chemical properties and hazard prediction of endpoints required to evaluate FCM substances. Moreover, the SILIFOOD application also provides information on the ED potential, an essential endpoint for FCM substances. In addition, the inclusion of the Cramer class model allows the application of the TTC approach when appropriate (for chemicals out of the exclusion criteria). The integration of these *in silico* models is particularly useful for evaluating data-poor chemicals such as non-evaluated FCM substances or NIAS. Importantly, since the prediction's reliability should always be carefully considered when interpreting the model's outcome, the level of reliability is indicated in the output report of the tool. By combining existing regulatory and toxicological data with hazard predictions from (Q)SAR models in a single report, the SILIFOOD tool can provide this information quickly and efficiently, supporting a fast risk assessment of FCM substances migrating into food. This was demonstrated in the four case studies.

CRedit authorship contribution statement

Camille Streef: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ilse Van Overmeire:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Gianluca Selvestrel:** Writing – review & editing, Supervision, Resources, Methodology, Data curation. **Alessandra Roncaglioni:** Writing – review & editing, Supervision, Resources, Methodology, Data curation. **Emilio Benfenati:** Writing – review & editing, Supervision, Resources, Methodology. **Alessio Sommavogio:** Writing – review & editing, Software, Resources, Methodology, Formal analysis, Data curation. **Alberto Mangano:** Writing – review & editing, Software, Resources, Methodology, Formal analysis, Data curation. **Serena Manganelli:** Writing – review & editing, Writing – original draft. **Els Van Hoeck:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Birgit Mertens:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ilse Van Overmeire reports financial support was provided by Federal Public Service Health Food Chain Safety and Environment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

The authors do not have permission to share data.

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